

AI System to Automatically Review and Summarize Research Papers

Statement of Purpose

The purpose of this project is to automate the initial stages of research paper review by building an AI-based system that searches, retrieves, and prepares academic papers for analysis and summarization, thereby reducing manual research effort.

Scope of Project

This project focuses on automated research paper retrieval, PDF handling, and preprocessing during the initial phase. The scope is limited to a small number of papers per topic and serves as a foundation for further analysis and summarization.

Methodology

- Step1: Identified the project requirements and finalized the research review workflow.
- Step 2: Set up the Python development environment and installed required libraries.
- Step3: Integrated the Semantic Scholar API to search research papers using a topic query.
- Step 4: Implemented automatic downloading of selected research papers in PDF format.
- Step 5: Performed initial text extraction from downloaded PDFs.
- Step 6: Organized extracted content for further analysis and summarization stages.

Week 1–2 Tasks

- Environment setup and dependency installation.
- Semantic Scholar API integration.
- Automated PDF download implementation.
- Initial text extraction from research papers.
- Dataset organization for analysis.

System Flow Diagram

User Input (Research Topic)
Paper Search (Semantic Scholar API)
PDF Download
Text Extraction &Preprocessing
Analysis & Key Findings Extraction
Cross-Paper Comparison
Summary & Draft Generation
Review &Refinement

Algorithm Flowchart

